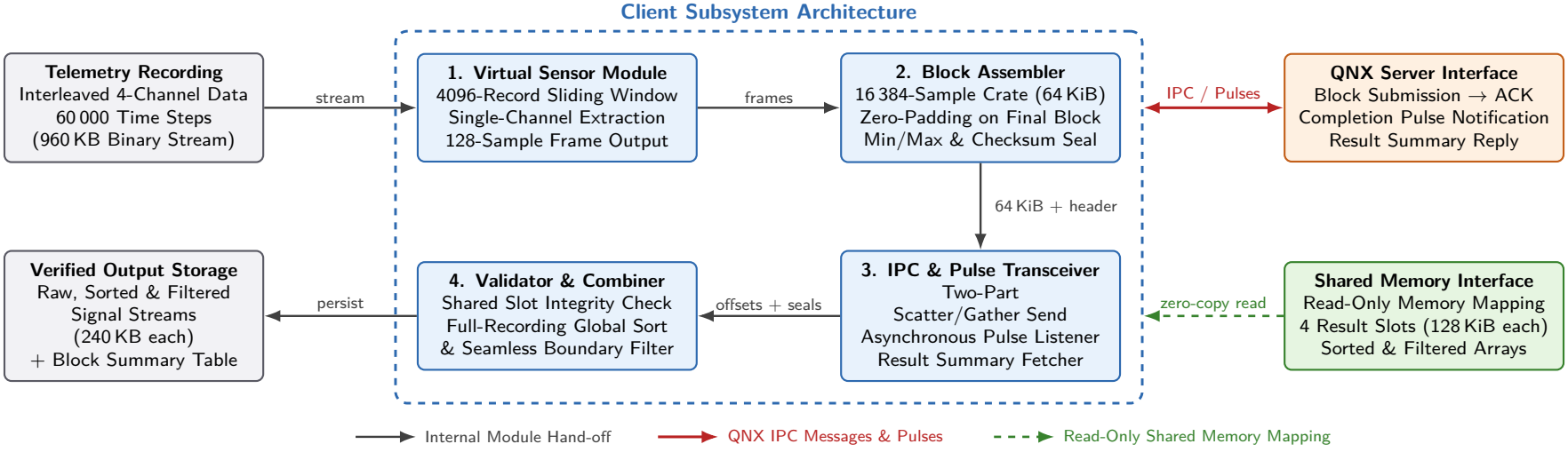

Client Subsystem Architecture
Telemetry Recording
Interleaved 4-Channel Data
60 000 Time Steps
(960 KB Binary Stream)
stream
1. Virtual Sensor Module
4096-Record Sliding Window
Single-Channel Extraction
128-Sample Frame Output
frames
2. Block Assembler
16 384-Sample Crate (64 KiB)
Zero-Padding on Final Block
Min/Max & Checksum Seal
IPC / Pulses
QNX Server Interface
Block Submission → ACK
Completion Pulse Notification
Result Summary Reply
64 KiB + header
Verified Output Storage
Raw, Sorted & Filtered
Signal Streams
(240 KB each)
+ Block Summary Table
persist
4. Validator & Combiner
Shared Slot Integrity Check
Full-Recording Global Sort
& Seamless Boundary Filter
offsets + seals
3. IPC & Pulse Transceiver
Two-Part
Scatter/Gather Send
Asynchronous Pulse Listener
Result Summary Fetcher
zero-copy read
Shared Memory Interface
Read-Only Memory Mapping
4 Result Slots (128 KiB each)
Sorted & Filtered Arrays
Internal Module Hand-off
QNX IPC Messages & Pulses
Read-Only Shared Memory Mapping